

06

CARBON MATRIX

Continuous 2/2 twill weave with selective reinforcement patches.

48.1 g

PLA · 100 % infill · solid

✓ UNDER 50 G IN ALL 4 MATERIALS



3/4 view · rendered from the STL, colours are preview only

AT A GLANCE

Overall height	132.82 mm
Max footprint (foot)	42.63 × 36.61 mm
Bore	32.06 × 26.04 mm, 127.8 deep
Textured band	z 21.5 – 116.0 mm (94.5 tall)
Outer size in band (max)	37.5 × 31.4 mm
Peak texture height	2.73 mm above bore
Base wall / min rim	1.2 mm / 0.8 mm
Volume / surface area	38.77 cm³ / 307 cm²

WEIGHT · 100 % INFILL

PLA	1.24 g/cm³		48.1 g
PETG	1.27 g/cm³		49.2 g
TPU	1.21 g/cm³		46.9 g
ABS	1.04 g/cm³		40.3 g

50 g limit

COLOURWAY



3MF print palette

PRINT PROFILE

Infill / layer height	100 % / 0.20 mm
Layers	665
Orientation	upright, butt end down
Supports	none required
Filament, 1.75 mm	≈ 16.1 m (100 % flow)

DESIGN INTENT

A woven-carbon look built as real geometry: every tow rises over and dives under its neighbours in a 2/2 twill. Smooth, slightly raised patches suggest selective reinforcement where the palm loads the handle.

The weave is the finest texture in the set (0.4 mm relief on a 2 mm pitch).

VERIFIED AGAINST THE REFERENCE

Bore vs reference	identical
Outer envelope vs reference	+0.001 mm at most
Rim vs reference	+0.40 mm (deliberate, ≥ 0.8 mm wall)
Bodies / enclosed voids	1 / none
Material under 0.4 mm wide	0.000 % of volume
Longest bridge	none

KEY FEATURES

- 60 tows around, 2.04 mm pitch, 2/2 twill
- Weave relief about 0.4 mm; every tow is a rounded ridge
- Smooth reinforcement patches stand about 0.5 mm above the weave

FILES

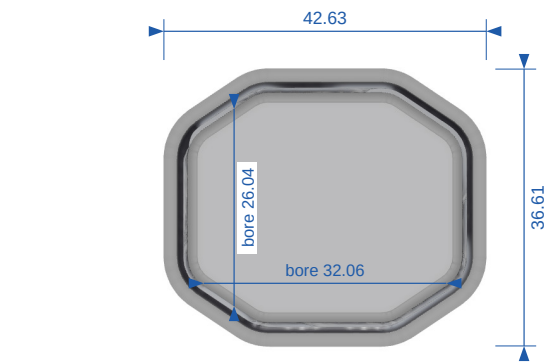
stl/06_carbon_matrix.stl	19.9 MB · 398,860 triangles
3mf/06_carbon_matrix.3mf	4.7 MB · colour

✓ FITMENT VERIFIED

✓ SOLID · ONE CLOSED BODY

✓ PRINT-READY · 0.20 MM LAYERS

✓ 3MF VALIDATED



TOP VIEW (from +Z)

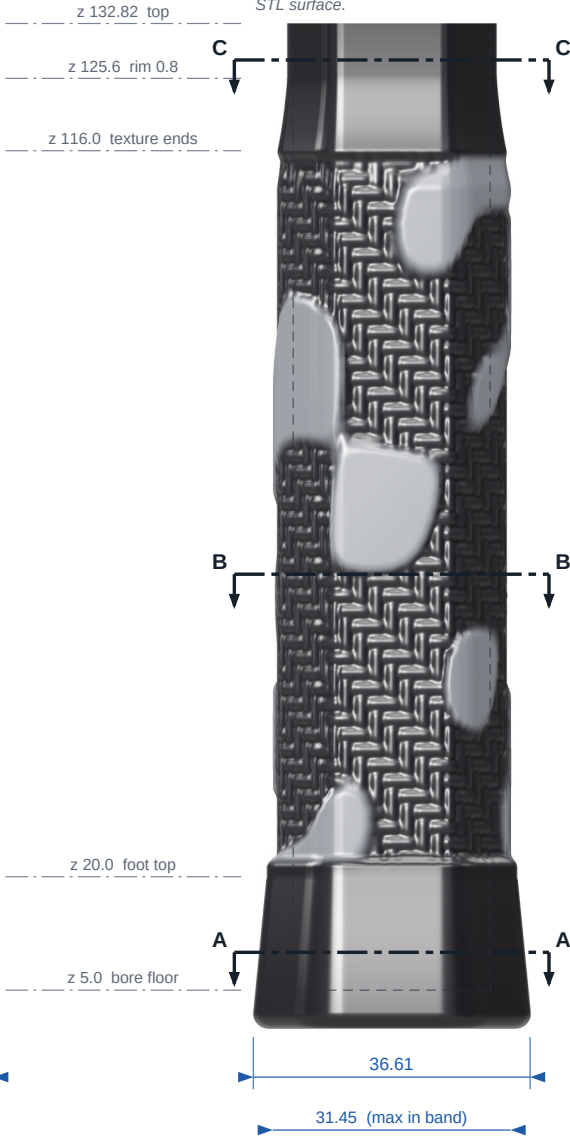


FRONT VIEW (from -Y)

KEY DIMENSIONS

Overall height	132.82
Overall width × depth (foot)	42.63 × 36.61
Bore width × depth	32.06 × 26.04
Bore depth (floor at z 5.0)	127.82
Outer W × D, textured band (max)	37.46 × 31.45
Foot top (z 20)	39.16 × 33.13
Butt face (z 0)	40.56 × 34.54
Peak texture height above bore	2.73
Base wall / minimum rim wall	1.20 / 0.80

All values in mm. Dashed lines: bore (hidden edges). Renders are the actual STL surface.



RIGHT VIEW (from +X)

SCALE 1 : 1



PART

CARBON MATRIX

Pickleball paddle grip sleeve, closed butt, open throat

GENERAL NOTES

- Views are true size: print this sheet at 100 %, do not fit to page.
- Dimensions are measured from the STL, not nominal.
- Bore, floor and outer envelope match the reference PickleballGrip_1.3mf.
- Rim is held at a minimum of 0.8 mm (reference tapers to 0.36 mm).

SCALE

1 : 1

MATERIAL

PLA / PETG / TPU

MASS (PLA)

48.1 g

REVISION

2

UNITS

millimetres

INFILL

100 %

VOLUME

38.77 cm³

DATE

2026-09-30

PROJECTION



LAYER HEIGHT

0.20 mm

SURFACE AREA

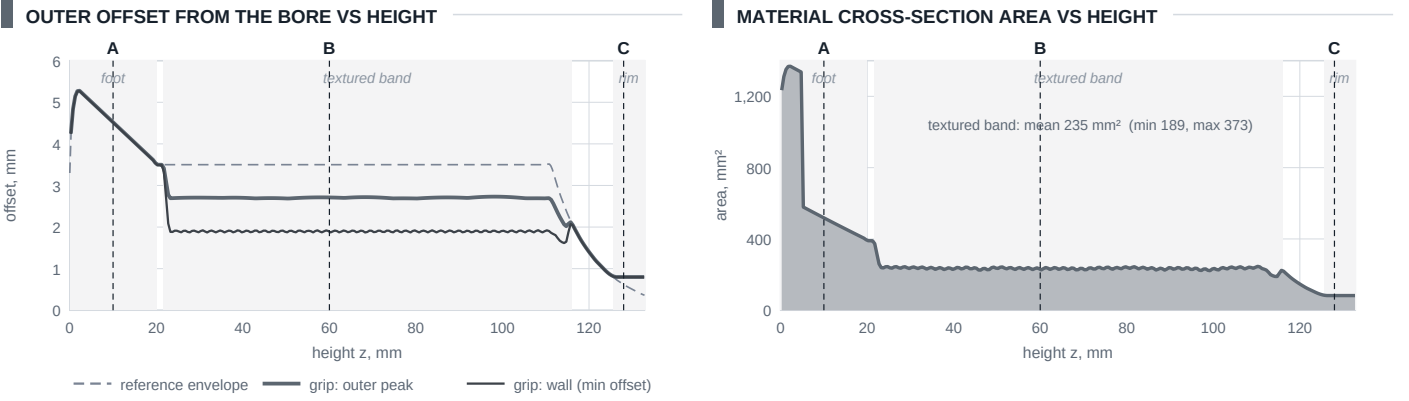
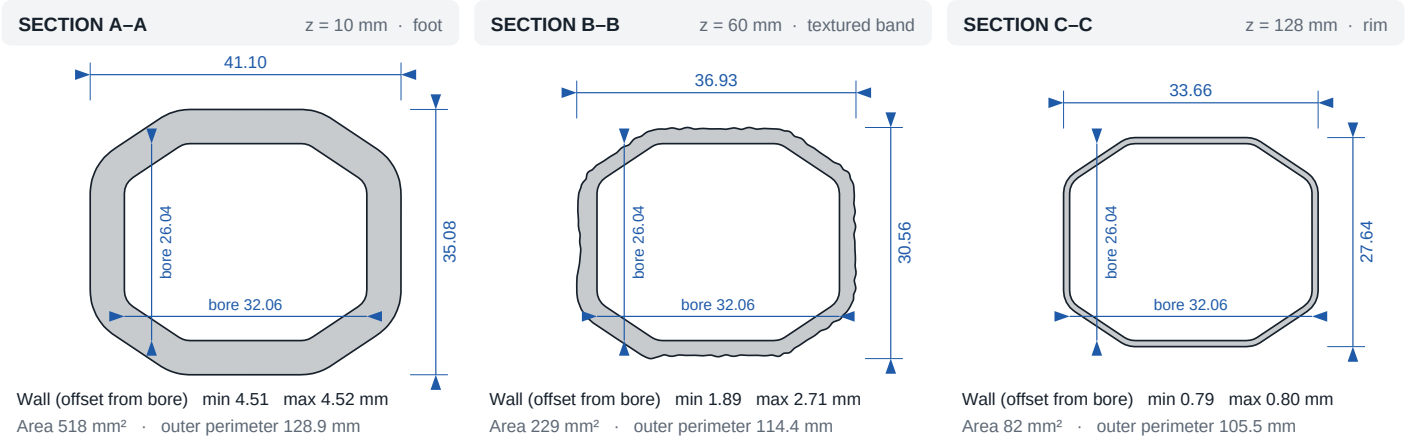
307 cm²

SHEET

2 of 4

Sections & profiles

True-size cross-sections, how the outer surface and material change with height, and the dimension tables.



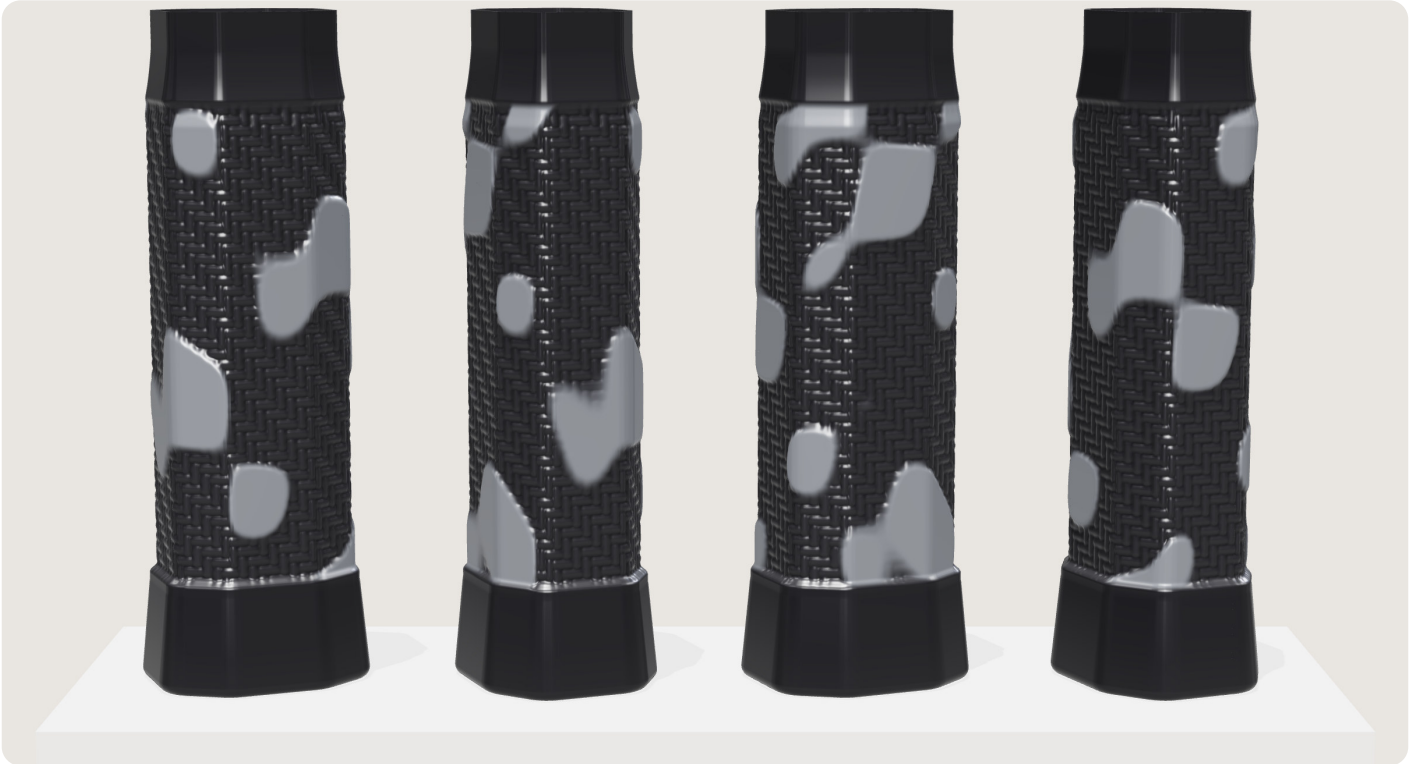
OVERALL & BORE	
Overall length (Z)	132.82 mm
Overall width (X)	42.63 mm
Overall depth (Y)	36.61 mm
Bore width × depth	32.06 × 26.04 mm
Bore area / perimeter	751.0 mm² / 100.5 mm
Bore depth (floor at z 5.0)	127.82 mm
Butt face (z 0)	40.56 × 34.54 mm
Outer W × D in band (max)	37.46 × 31.45 mm
Circumference in band	113.7 – 115.2 mm
Volume / surface area	38.77 cm³ / 306.6 cm²

OUTER DIMENSIONS BY HEIGHT

Z (MM)	OUTER W	OUTER D	PERIMETER	AREA MM²	WALL MIN	PEAK OFFSET
5.0	42.08	36.06	132.0	958	5.00	5.01
10.0	41.10	35.08	128.9	518	4.51	4.52
20.0	39.11	33.09	122.6	393	3.52	3.52
30.0	36.88	30.88	114.8	237	1.89	2.71
45.0	37.26	31.01	114.5	237	1.91	2.70
60.0	36.96	30.56	114.5	230	1.89	2.71
75.0	36.88	30.87	114.5	237	1.89	2.69
90.0	36.78	31.08	114.9	238	1.91	2.71
105.0	36.89	30.87	114.7	236	1.88	2.70
116.0	36.21	30.20	113.5	221	2.06	2.08
125.0	33.77	27.75	105.8	88	0.85	0.85
132.0	33.66	27.64	105.5	82	0.79	0.80

Views & surface

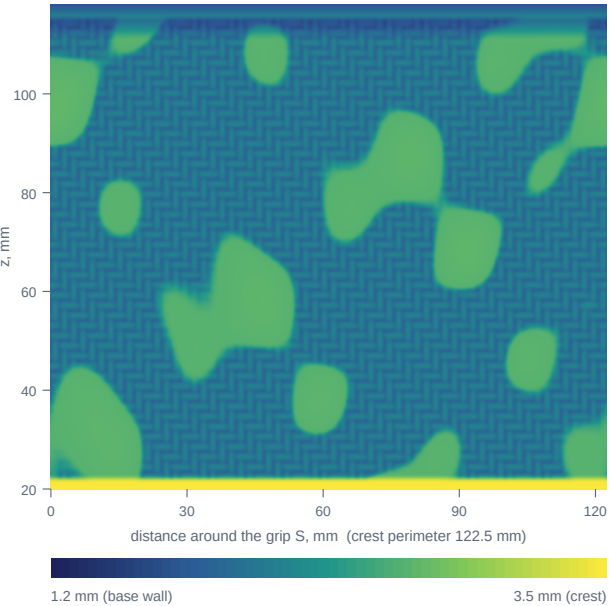
Four-way turntable, close-ups of the surface, and the unrolled height map with the design parameters.



Four views, 90° apart · rendered from the STL, colours are preview only



UNROLLED SURFACE · HEIGHT ABOVE BORE



DESIGN PARAMETERS

Tows around	60 (2/2 twill)
Tow pitch	2.04 mm
Weave heights	1.7 – 2.2 mm above bore
Patch heights	2.3 – 2.7 mm above bore
Patch coverage	about 23% of the band
Weave / patch colours	carbon / grey
Textured band	94.5 mm tall

PRINT & VERIFICATION

Result: fitment / print-ready	PASS / PASS
Bodies · genus	1 · 0
Bore vs reference	identical
Envelope over reference	+0.001 mm
Material under 0.4 mm wide	0.000 %
Longest bridge	none